

KEERTHANA THUMMALA

Software Engineer | Full-Stack Developer | AI/ML | Data Analysis
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SUMMARY

Software Engineering student (B.S., expected 2027) with production experience across full-stack development, AI agent systems, and data analysis. Comfortable building end-to-end, from a React front end to a backend API to a retrieval or classification pipeline behind it, and uses AI-assisted development tools as a normal part of the workflow, not a novelty. Contributed to a 20% reduction in infrastructure costs and a 25% improvement in delivery velocity through CI/CD automation in a production environment.

TECHNICAL SKILLS

- **Languages:** Python, Java (OOP), JavaScript, TypeScript, SQL
- **Frontend & Backend:** React.js, Tailwind CSS, HTML5/CSS3, Node.js, FastAPI, RESTful APIs
- **Databases:** MySQL, MongoDB, ChromaDB (Vector DB)
- **AI / ML:** LangGraph, LangChain, RAG, Prompt Engineering, Ollama, Scikit-learn, Pandas, NumPy, PCA
- **Data Analysis:** Exploratory Data Analysis, Feature Engineering, Matplotlib/Seaborn, Model Evaluation
- **AI-Assisted Development:** Claude Code, GitHub Copilot, Gemini, ChatGPT, Lovable, Canva, DeepSeek, LLM API integration, prompt engineering for coding workflows
- **Tools & Methodologies:** Git/GitHub, Agile/Scrum, Jira, SDLC, System Design

WORK EXPERIENCE

Full Stack Developer | [RCS Tech LLP](#) | 12/25 - 04/26

- Engineered scalable web architecture for the TRAXX platform, reducing infrastructure expenses by 20%.
- Streamlined development workflows and CI/CD pipelines, improving team engineering efficiency by 25%.
- Led sprint planning and code reviews in an Agile framework to keep delivery aligned with client specifications.

Software Development Intern | CodSoft | 01/25 - 02/25

- Built modular applications in Java and Python, improving code reusability and system scalability.
- Refined application logic and memory management, increasing execution efficiency by 25%.

MAJOR PROJECTS

OrbitOps Assistant | *Python, LangGraph, LangChain, ChromaDB, BM25, Sentence-Transformers, Ollama, FastAPI, React*

- Merged a document-grounded RAG pipeline with a multi-tool LangGraph agent into one deployable assistant that both answers questions from uploaded documents with cited sources and pulls live data from external APIs.
- RAG side: hybrid retrieval (BM25 + dense vector search via ChromaDB, fused with reciprocal rank fusion) across 4 retrieval methods, benchmarked with Precision, Recall, Hit Rate, and MRR, with grounded answering that reports insufficient information instead of fabricating.
- Agent side: a multi-step decision graph that calls live APIs, combines results, and generates structured answers, with guardrails against empty input and prompt-injection, scored by an automated LLM-judge evaluation suite (10+ test cases).
- Shipped as a live, interactive full-stack web app (FastAPI backend, React frontend, local Ollama model), built end-to-end and deployed with zero paid resources.

Nutrient Profile Classification — Data Analysis | *Python, Pandas, Scikit-learn, Matplotlib/Seaborn, PCA*

- Cleaned a messy 1,590-item, 272-column nutrition dataset into an analysis-ready format.
- Engineered features (protein-per-calorie, sodium-to-potassium ratio) and built KNN and Random Forest classifiers.
- Identified a data leakage issue inflating model accuracy and proposed a corrected feature set.

Resume Tracker Extension | *React.js, Java, JSON, HTML/CSS*

- Built a browser extension with secure user registration/login and a dynamic job-application tracking system, using AI-assisted coding tools to accelerate development and debugging.
- Used JSON to persist and reload user data seamlessly across sessions.

EDUCATION

Bachelor of Science in Software Engineering | [The University of Queensland](#) | 2023 - 2027

Relevant coursework: Full-Stack Development, Data Structures & Algorithms, Agile Methodologies, Machine Learning, Data Science, LLMs.

RESEARCH

Honours Thesis: Benchmarking Hardness in Package-to-Truck Allocation for Near-Term Quantum Optimisation

Supervisor: Prof. Eleanor Rieffel | The University of Queensland | 2026

- Designed a benchmark suite of package-to-truck allocation problems (100–10,000 packages) grounded in real Australian data, with a hardness metric validated against solver statistics; ranked instances via OR-Tools CP-SAT, scoping future quantum work (QAOA) on the hardest set.

Tools: Python, OR-Tools, Qiskit, python-docx